12-24 第二篇

Self-Driving Cars: The Road Ahead

Self-driving cars, also called autonomous vehicles, are designed to navigate roads without a human driver. They use cameras, radar, lidar, and powerful computers to sense their surroundings, read traffic signs, and make decisions in real time. Companies and researchers have been testing these vehicles for years, and some are already on the road in limited areas. Supporters say they could reduce accidents caused by human error, ease traffic, and give mobility to people who cannot drive. [1]

How do they work? Sensors continuously scan the environment—other cars, pedestrians, cyclists, and obstacles. Software processes this data to build a 3D map of the world and plan a safe path. The car accelerates, brakes, and steers on its own, though many current systems still require a human to take over in complex situations. Levels of autonomy range from basic assistance—like cruise control—to full self-driving with no human input. [2]

Challenges remain. Who is responsible if a self-driving car causes an accident—the owner, the manufacturer, or the software developer? How should the car be programmed to choose between bad outcomes, such as hitting a pedestrian or swerving into another car? Bad weather, unclear road markings, and unexpected situations can confuse the sensors. These ethical and technical questions must be answered before self-driving cars become widespread. [3]

Regulation is evolving. Some countries have allowed testing on public roads with strict rules. Insurance and liability laws are being updated. As technology improves, more cities may see self-driving taxis or delivery vehicles. The transition from human drivers to autonomous systems will likely happen gradually, with mixed traffic—human and computer-driven—for a long time. [4]

Self-driving cars represent a major shift in how we move. Whether they will deliver on their promise of safer, more efficient travel depends on how well we address the technical, ethical, and legal questions ahead. [5]

选项

A. Self-driving cars hold the promise of transforming transportation, but realizing that promise requires careful attention to safety, ethics, and law. B. This technology relies on a combination of hardware and software working together to mimic—and perhaps surpass—human driving abilities. C. These questions show that developing self-driving cars is not only an engineering problem but also a deeply ethical one. D. For this reason, the rollout of autonomous vehicles will be gradual, with ongoing updates to rules and

technology. E. The future of driving may look different from today, but getting there will require solving hard problems along the way. F. For example, Tesla's Autopilot and Full Self-Driving features offer varying levels of driver assistance, though they still require the driver to pay attention.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：空前介绍自动驾驶汽车无需人类驾驶、使用摄像头雷达激光雷达与计算机感知环境做出决策、已有测试与有限上路、支持者称可减少人为事故缓解拥堵、为无法驾驶者提供出行，空后需总结无人驾驶的承诺与实现条件。A的"hold the promise of transforming transportation"与前文呼应，"realizing that promise requires careful attention to safety, ethics, and law"与前文对应。
- **[2] B**：
  - **线索**：空前说工作原理、传感器持续扫描、软件处理数据建3D图规划路径、自动加速刹车转向、当前多数系统复杂情况仍需人接管、 autonomy 层级从基础辅助到完全无人，空后需总结硬件软件协同模拟或超越人类驾驶。B的"This technology relies on a combination of hardware and software"指代前文，"mimic—and perhaps surpass—human driving abilities"与前文呼应。
- **[3] C**：
  - **线索**：空前说仍有挑战、事故责任归属、编程如何在不利后果间选择、恶劣天气模糊标线意外情况可干扰传感器、伦理与技术问题须在普及前回答，空后需总结开发不止是工程问题也是伦理问题。C的"These questions show"承接前文，"not only an engineering problem but also a deeply ethical one"与前文对应。
- **[4] D**：
  - **线索**：空前说监管在演变、有的国家允许严格规则下的公共道路测试、保险与责任法在更新、技术改进后或见自动驾驶出租车或配送车、从人类驾驶到自主系统的过渡可能渐进、人机混行将长期存在，空后需总结 rollout 的渐进性与规则技术更新。D的"For this reason"承接前文，"gradual, with ongoing updates to rules and technology"与前文呼应。
- **[5] E**：
  - **线索**：空前说无人驾驶代表出行方式的重大转变、能否实现更安全高效出行取决于我们如何应对技术伦理法律问题，空后需总结未来与路径上的难题。E的"The

future of driving may look different"与前文对应，"getting there will require solving hard problems"与前文呼应。

- **干扰项分析**：
  - **F. For example, Tesla's Autopilot and Full Self-Driving...**：以For example开头，为具体产品案例。各空均为概括或过渡句；[2]易被误选因涉及具体系统，但该空需总结硬件软件协同的工作原理，而非补充某品牌功能的细节。